

LECTURE 1

PROBABILITY

Predictive Market Modeling



Probability and its basic Terminology

EXPERIMENT

An action or process that leads to one of many possible outcomes, such as flipping a coin or rolling a die.

SAMPLE SPACE(Ω)

The set of all possible outcomes of an experiment; for example, in a dice roll, the sample space is $\{1, 2, 3, 4, 5, 6\}$.

EVENT

A subset of the sample space, which can be a single outcome or a group of outcomes, such as rolling an even number $\{2, 4, 6\}$.

Probability and its basic Terminology

MUTUALLY EXCLUSIVE

**EVENT A AND B
ARE MUTUALLY
EXCLUSIVE IF:**

$$P(A \cap B) = 0$$

INDEPENDENT

**EVENTS A AND B
ARE INDEPENDENT
IF:**

$$P(A \cap B) = P(A) \times P(B)$$

MUTUALLY EXHAUSTIVE

**EVENTS A AND B
ARE MUTUALLY
EXHAUSTIVE IF:**

$$P(A \cup B) = 1$$

CONCEPT OF RANDOM VARIABLE

A real-valued function $X: \Omega \rightarrow \mathbb{R}$ defined on the sample space Ω is called a random variable.

Example: Flipping a coin

$\Omega = \{\text{Heads (H), Tails (T)}\}$.

$X(H) = 1$ (Heads maps to 1),

$X(T) = 0$ (Tails maps to 0).

$P(X=1)$ is the probability of flipping Heads.

$P(X=0)$ is the probability of flipping Tails.

**This is an example
of Discrete Random
Variable**

Distribution Function of a Random Variable

Cumulative Distribution Function (CDF):

The Cumulative Distribution Function of a random variable X is defined as:

$$F_X(x) = P(X \leq x),$$

where $F_X(x)$ gives the probability that the random variable X takes a value less than or equal to x .

$$\Omega = \{HH, HT, TH, TT\}.$$

$$X \in \{0, 1, 2\}.$$

$$X(TT) = 0 \text{ (No Heads),}$$

$$X(HT) = 1 \text{ (One Head),}$$

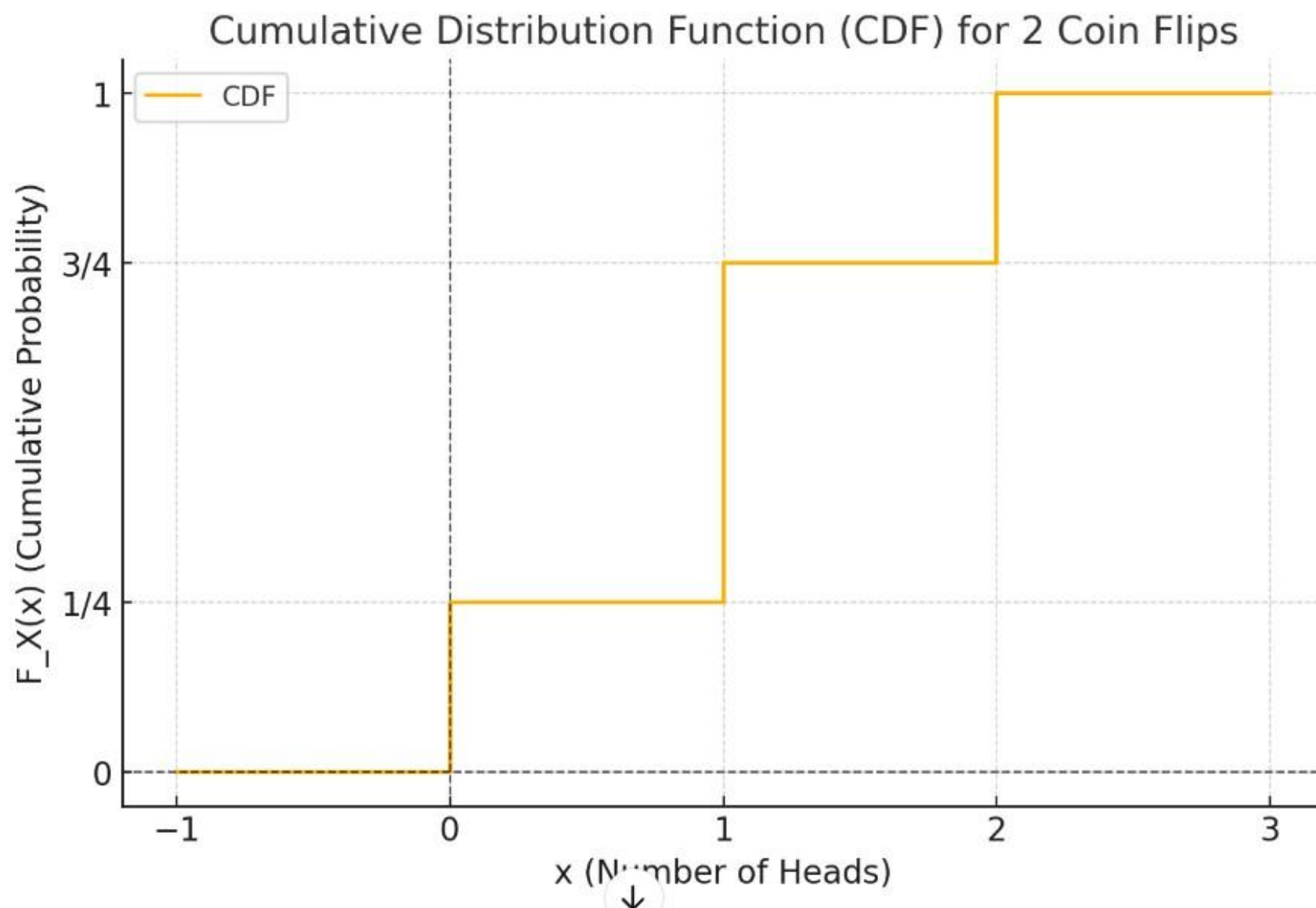
$$X(TH) = 1 \text{ (One Head),}$$

$$X(HH) = 2 \text{ (Two Heads).}$$

- $P(X = 0) = P(TT) = \frac{1}{4},$
- $P(X = 1) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2},$
- $P(X = 2) = P(HH) = \frac{1}{4}.$

Distribution Function of a Random Variable

Cumulative Distribution Function (CDF):

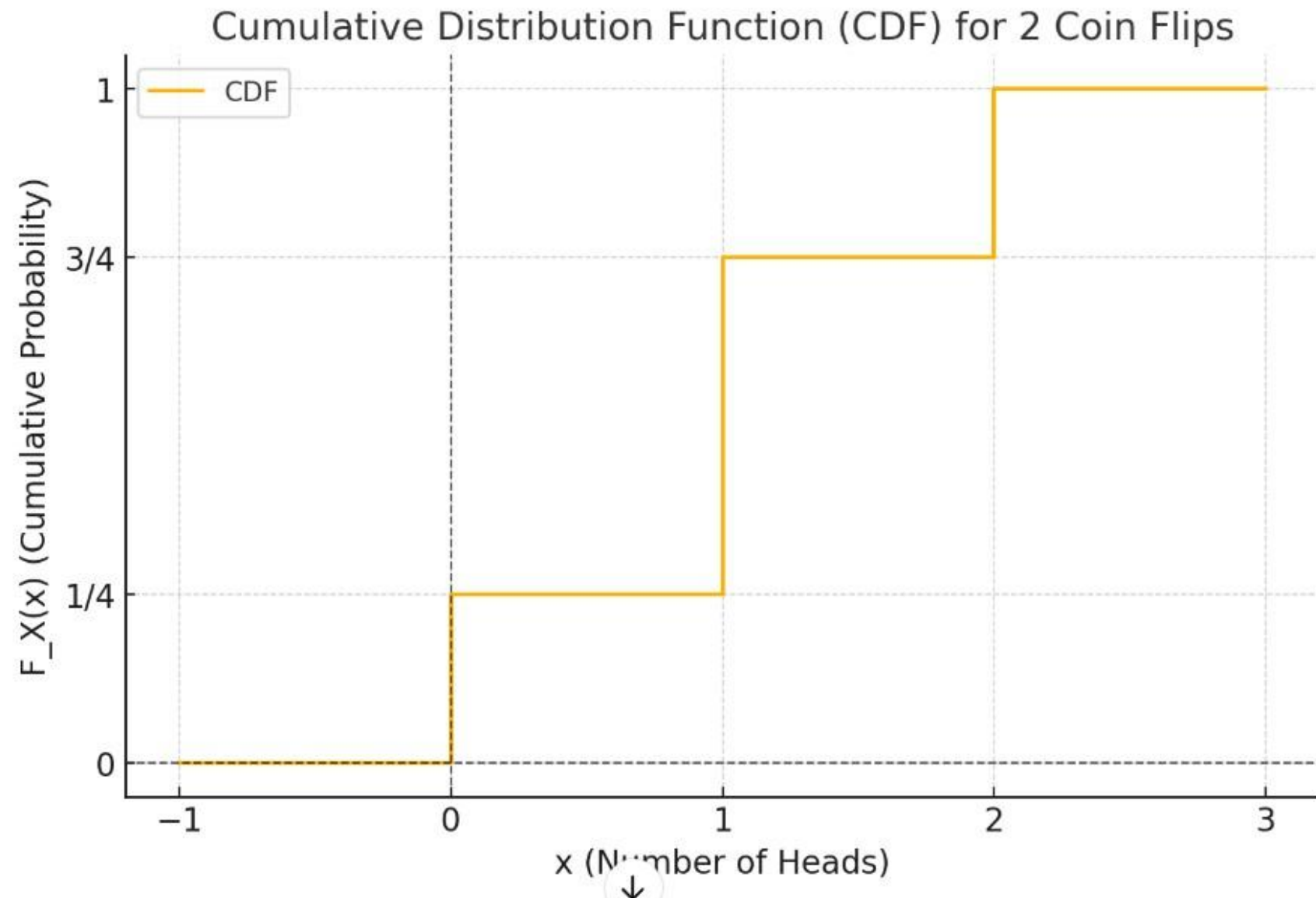


- $P(X = 0) = P(\text{TT}) = \frac{1}{4},$
- $P(X = 1) = P(\text{HT}) + P(\text{TH}) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2},$
- $P(X = 2) = P(\text{HH}) = \frac{1}{4}.$

$$F_X(x) = P(X \leq x).$$

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0, \\ \frac{1}{4} & \text{if } 0 \leq x < 1, \\ \frac{3}{4} & \text{if } 1 \leq x < 2, \\ 1 & \text{if } x \geq 2. \end{cases}$$

Properties of Cumulative Distribution Function (CDF):



$$F_X(x_1) \leq F_X(x_2), \quad \text{for } x_1 < x_2.$$

$$0 \leq F_X(x) \leq 1.$$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

$$P(X = x) = F_X(x) - F_X(x^-),$$

$\lim_{x \rightarrow -\infty} F_X(x) = 0$: As x approaches $-\infty$, the CDF approaches 0.

$\lim_{x \rightarrow \infty} F_X(x) = 1$: As x approaches ∞ , the CDF approaches 1.

PMF(Probability mass Function) (defined for discrete random variable)

Described by a Probability Mass Function (PMF), $P(X = x)$, which gives the probability of each value:

$$\sum_x P(X = x) = 1.$$

PDF(Probability density Function) (defined for continuous random variable)

Continuous Random Variable X
Takes any value within a range (e.g., [a,b] or R).

PDF(Probability density Function) (defined for continuous random variable)

The Cumulative Distribution Function (CDF) for a continuous random variable X is defined as:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt,$$

where $f_X(t)$ is the Probability Density Function (PDF).

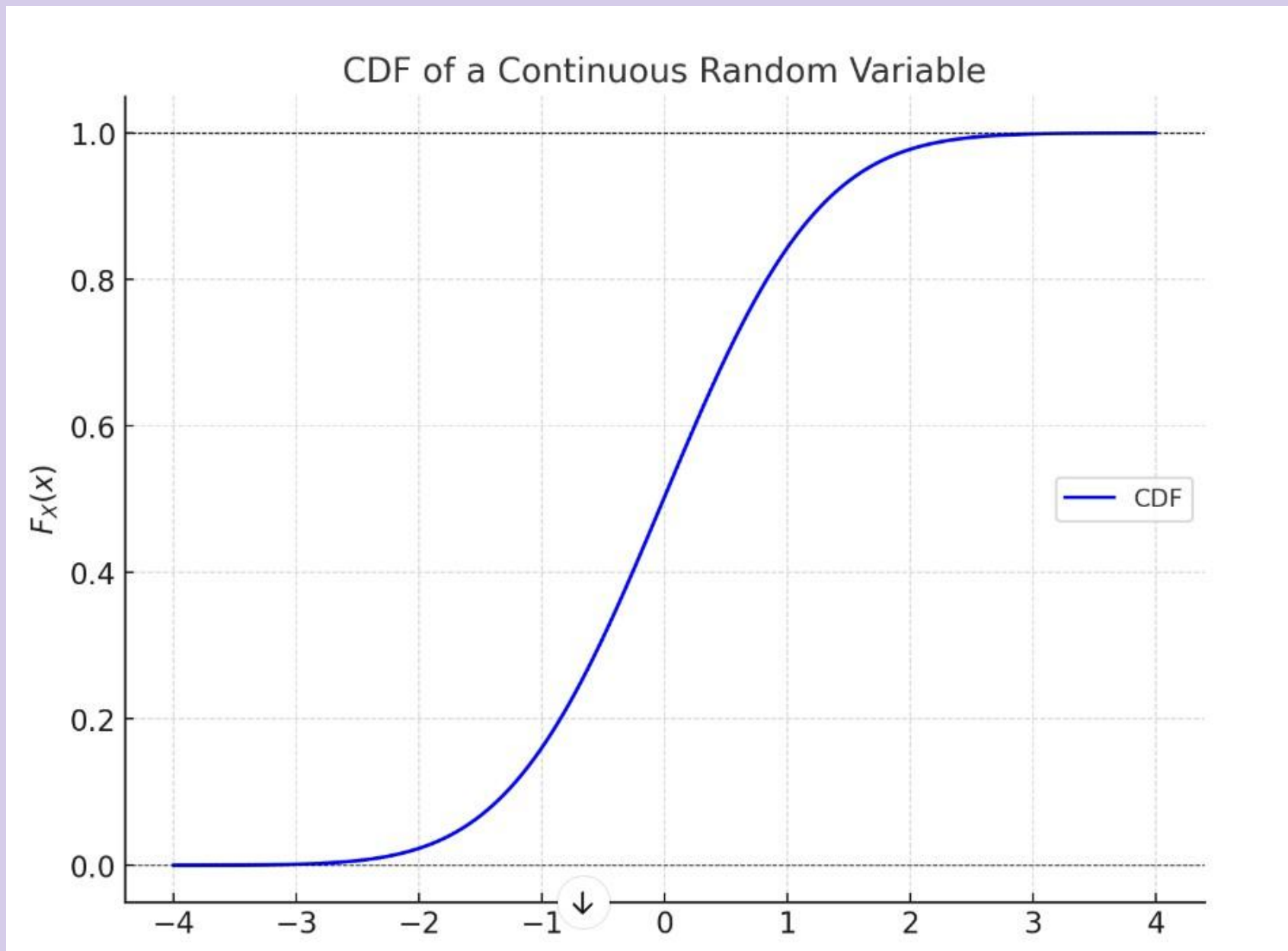
Discrete R.V.

$$F_X(x) = P(X \leq x) = \sum_{-\infty}^x \text{Pmf}(x)$$

Continuous R.V.

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt,$$

CDF for continuous Random Variable



Note that all the properties of Cdf are held true here.

Expectation Value for Discrete Random Variable

The **expectation** (mean) of a discrete random variable X is the weighted average of its possible values, defined as:

$$\mu = \mathbb{E}[X] = \sum_x x \cdot P(X = x),$$

The **variance** measures the spread of X around the mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 \cdot P(X = x).$$

An equivalent formula is:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Expectation Value for Discrete Random Variable

The **expectation** (mean) of a continuous random variable X is defined as:

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx,$$

The **variance** for continuous X is:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f_X(x) dx.$$

Alternatively:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Standard Distributions

Bernoulli Distribution

- **Definition:** Models a single trial with two outcomes: success (1) or failure (0).
- **PMF:**

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

where p is the probability of success.

- **Mean:** $\mu = p$
- **Variance:** $\sigma^2 = p(1 - p)$
- **Example:** Flipping a coin (success = heads).

Standard Distributions

Binomial distribution

- **Definition:** Models the number of successes in n independent Bernoulli trials.
- **PMF:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

where n is the number of trials and p is the probability of success.

- **Mean:** $\mu = n \cdot p$
- **Variance:** $\sigma^2 = n \cdot p \cdot (1 - p)$
- **Example:** Number of heads in 10 coin flips.

Standard Distributions

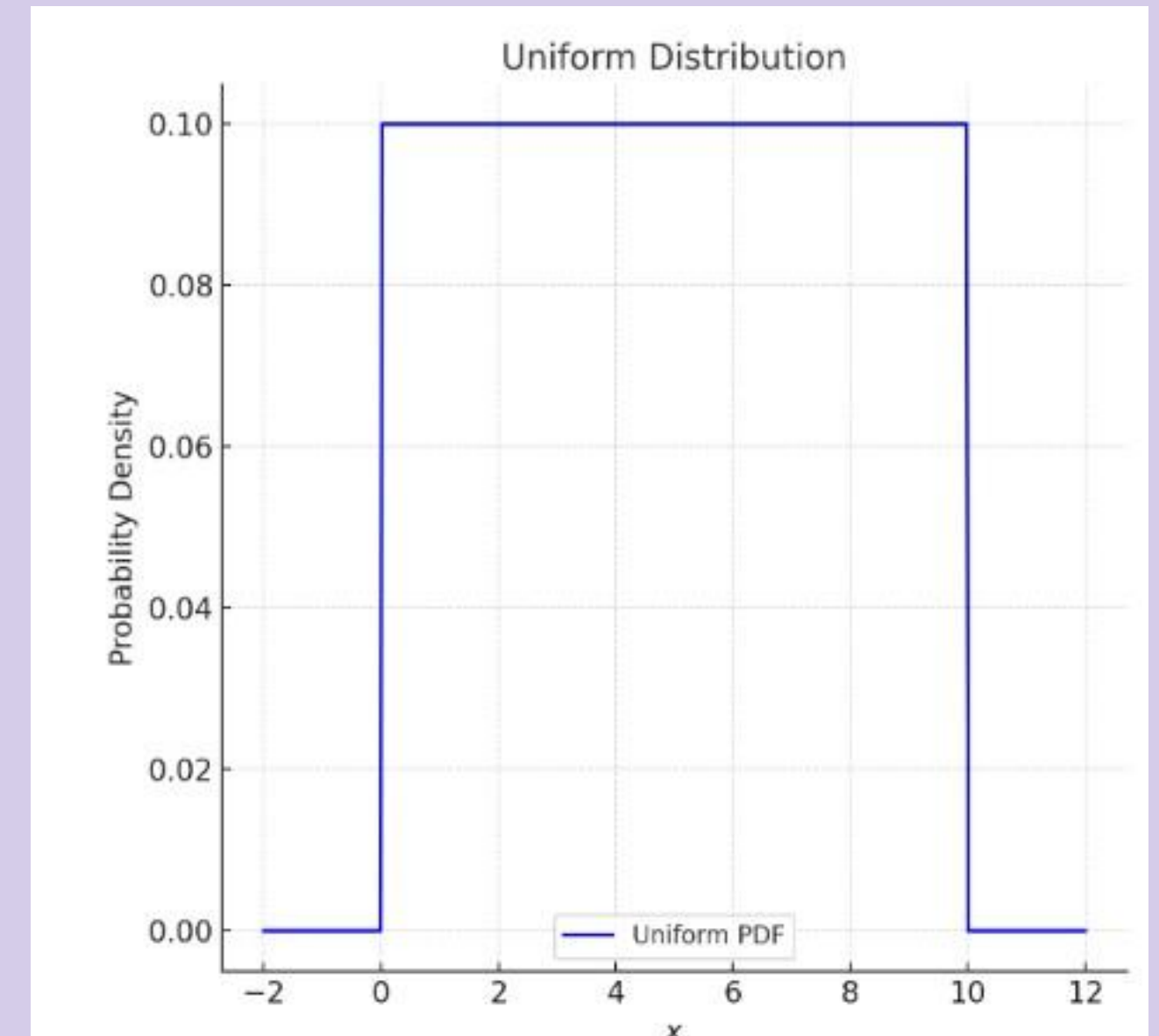
Uniform distribution

- **Definition:** Models a random variable equally likely to take any value in the interval $[a, b]$.

- PDF:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- **Mean:** $\mu = \frac{a+b}{2}$
- **Variance:** $\sigma^2 = \frac{(b-a)^2}{12}$
- **Example:** Randomly picking a number between 1 and 10.



Standard Distributions

Normal Distribution

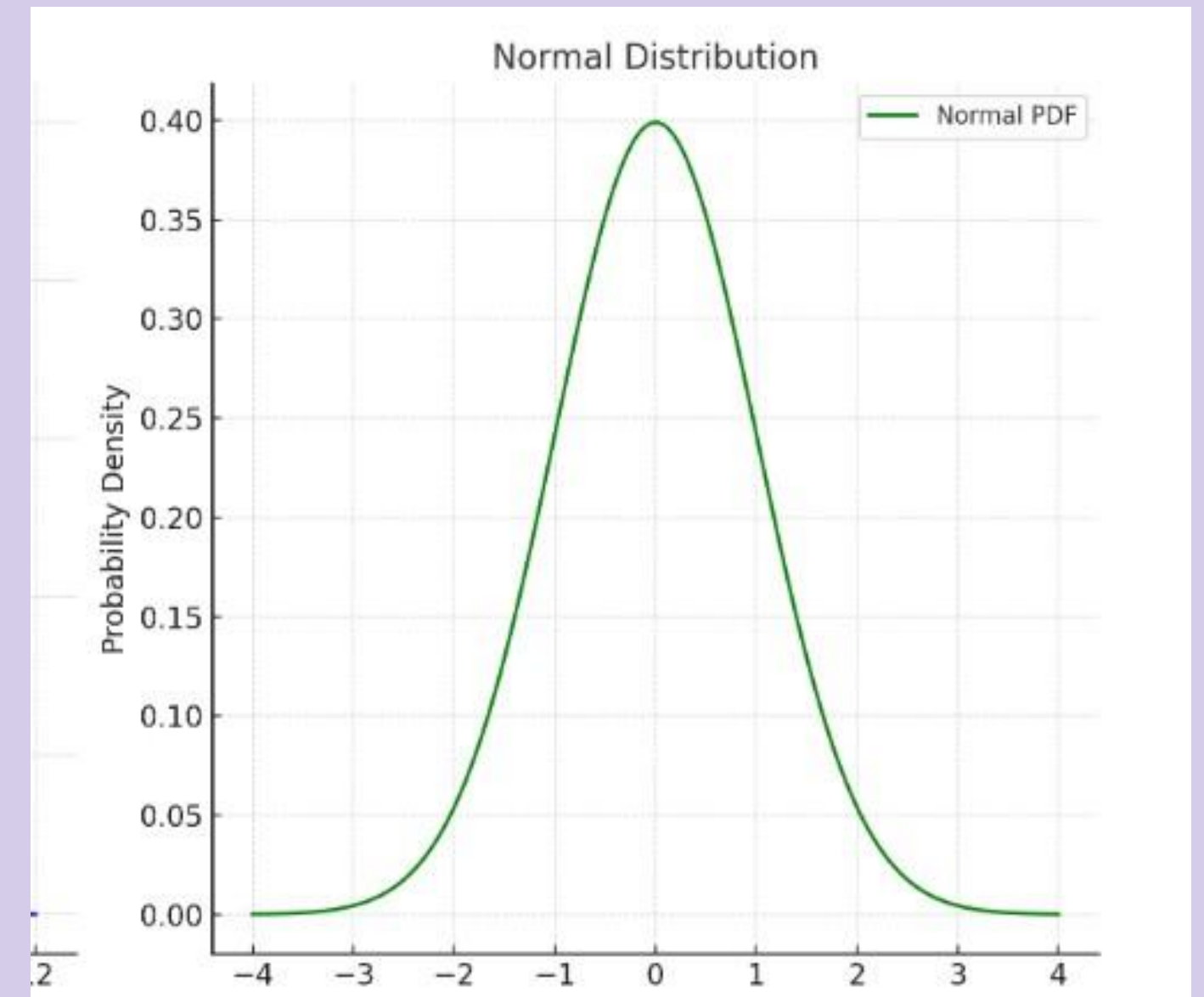
- **Definition:** Models a symmetric, bell-shaped distribution.
- **PDF:**

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is the mean and σ^2 is the variance.

- **Key Properties:**
 - Symmetric around μ .

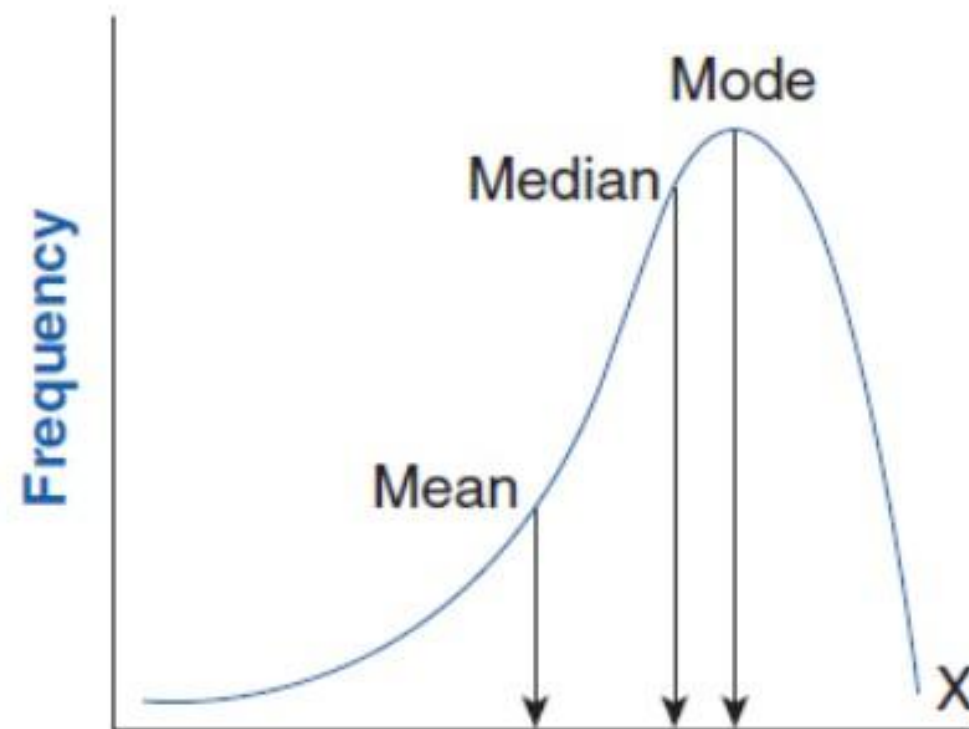
If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.



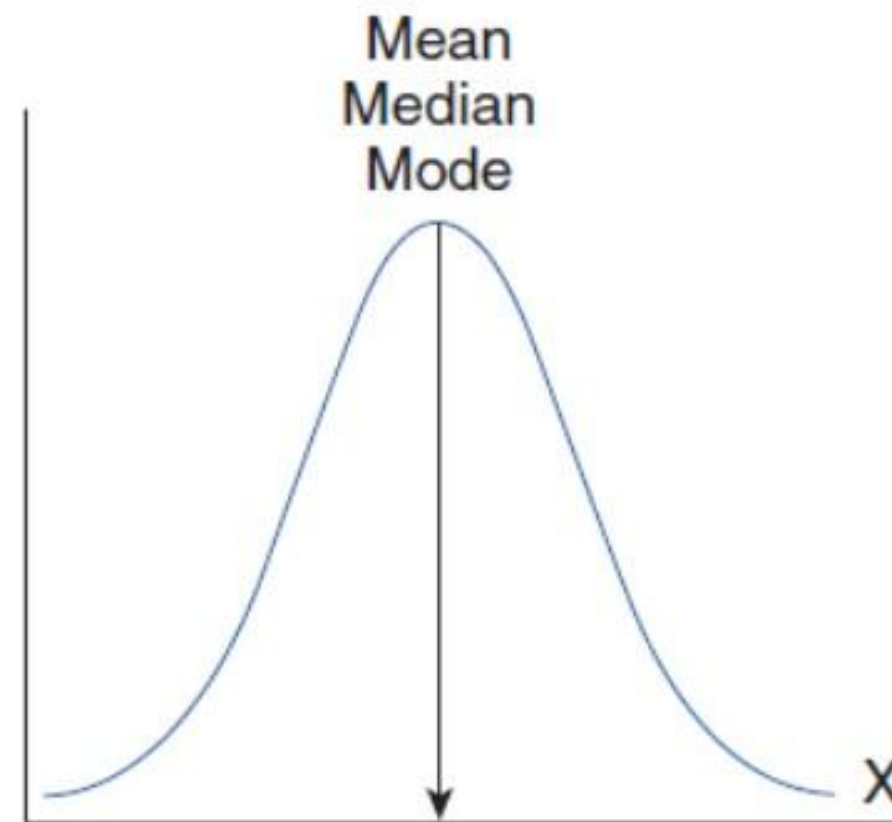
Measure of Skewness

$$\text{Skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right)^{3/2}}$$

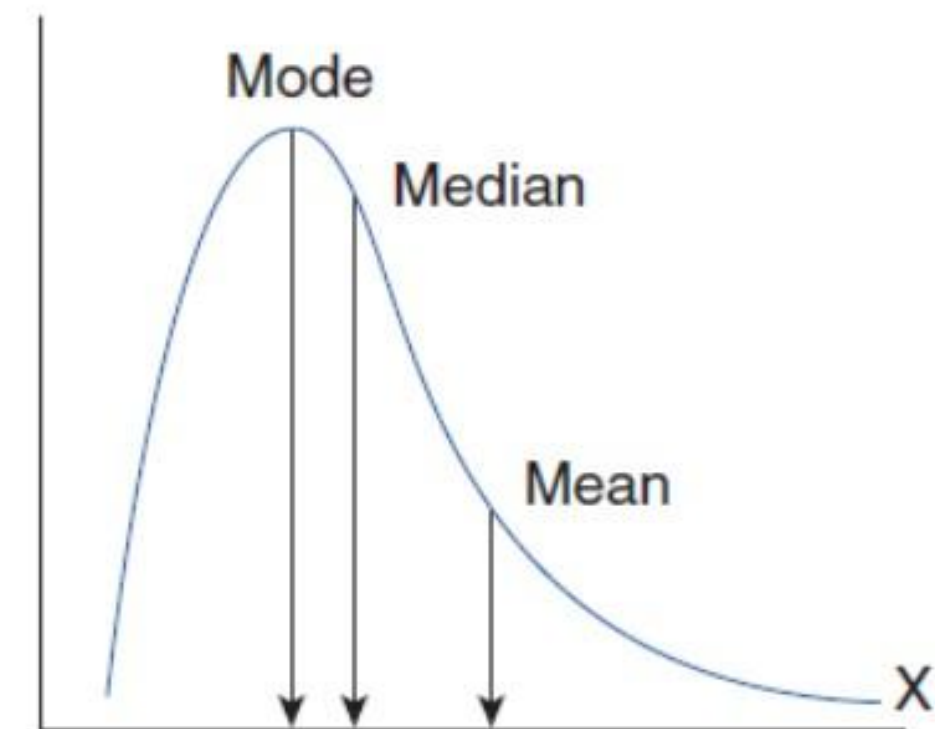
(a) Negatively skewed



(b) Normal (no skew)



(c) Positively skewed



Co - variance

Measures the degree to which two random variables vary together.

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Correlation

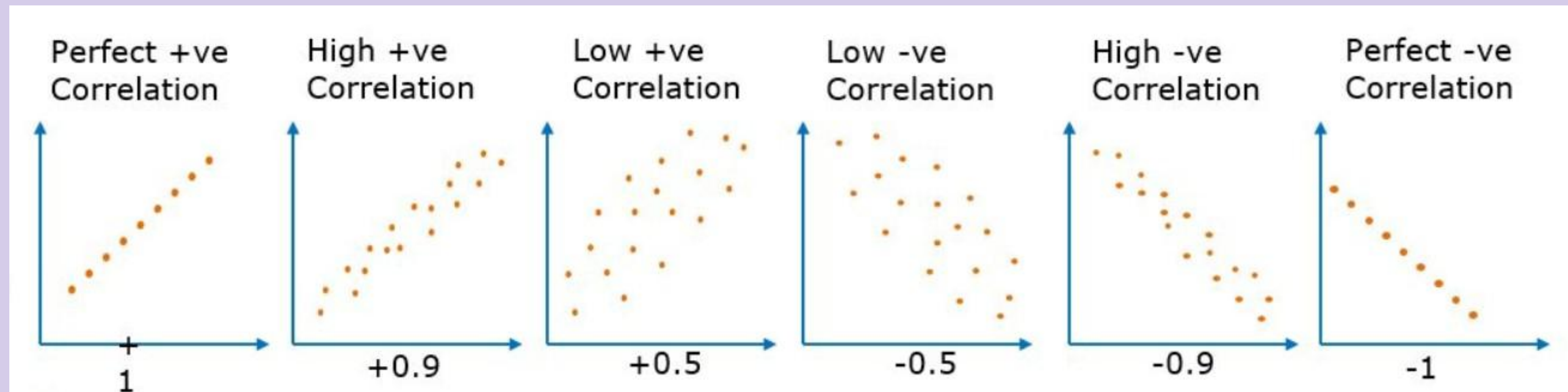
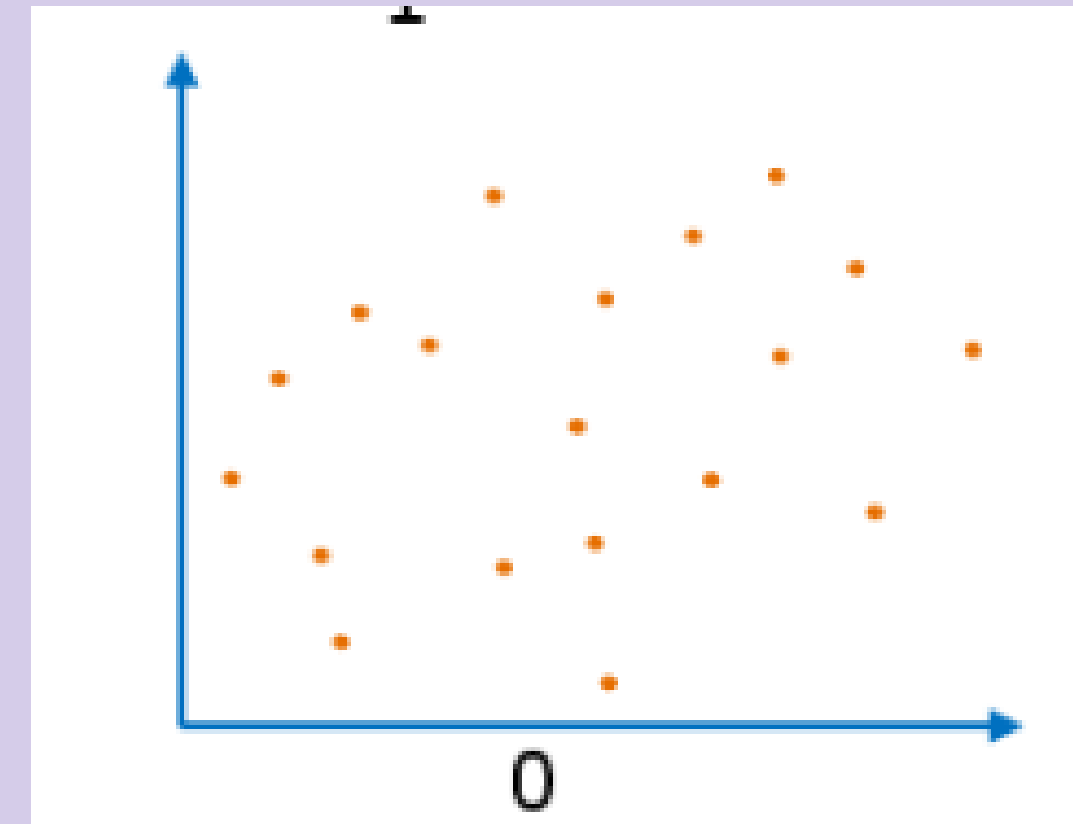
A standardized measure of the strength and direction of the linear relationship between two variables.

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$r = 1$: Perfect positive linear relationship.

$r = -1$: Perfect negative linear relationship.

$r = 0$: No linear relationship.



Joint Distributions

A joint distribution describes the probability of two or more random variables occurring simultaneously. It can be for discrete or continuous random variables.

Discrete Joint Distribution:

- For random variables X and Y :

$$P(X = x, Y = y) = \text{Joint Probability Mass Function (PMF)}.$$

- Example: Tossing two dice, where X and Y represent the outcomes of each die.

Joint Distributions

Continuous Joint Distribution:

- Described using a **Joint Probability Density Function (PDF)** $f(x, y)$, such that:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx.$$

- Example: Heights and weights of individuals.

Marginal Distribution

Found by summing (discrete case) or integrating (continuous case) the joint distribution over the other variable.

- Discrete:

$$P(X = x) = \sum_y P(X = x, Y = y).$$

- Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

Conditional Distribution

The probability of one variable given the value of another.

- Formula:

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} \quad (\text{Discrete}).$$

$$f(X \mid Y) = \frac{f(x, y)}{f_Y(y)} \quad (\text{Continuous}).$$

X and Y are independent if:

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y) \quad (\text{Discrete}).$$

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad (\text{Continuous}).$$

THANK YOU :))